

Web Programming Final Project

Bookstore

Personal Reading App with AI Assistant



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The Problem



Fragmented Experience

Reading books online is fragmented — different apps, no sync, no notes



Cost Barrier

Most e-readers are paid or require account on external services



No Intelligence

No way to ask questions about what you're reading



Lost Progress

Readers lose their place, highlights, and notes

→ Current solutions fail to provide a unified, intelligent reading experience



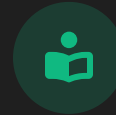
Our Solution

Bookstore



Private Upload

Upload any PDF or EPUB — it's stored privately, just for you



Clean Reader

Read in browser with a clean, distraction-free reader



Notes & Highlights

Notes & highlights saved per page



AI Assistant

Built-in AI assistant answers questions about your book



Reading Statistics

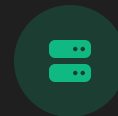
Reading statistics and daily goals to track your progress

Core Features



Reader

- ✓ PDF and EPUB support
- ✓ Scroll and paged reading modes
- ✓ Auto-scroll with adjustable speed
- ✓ Light, dark, and sepia themes
- ✓ Font, size, line spacing controls

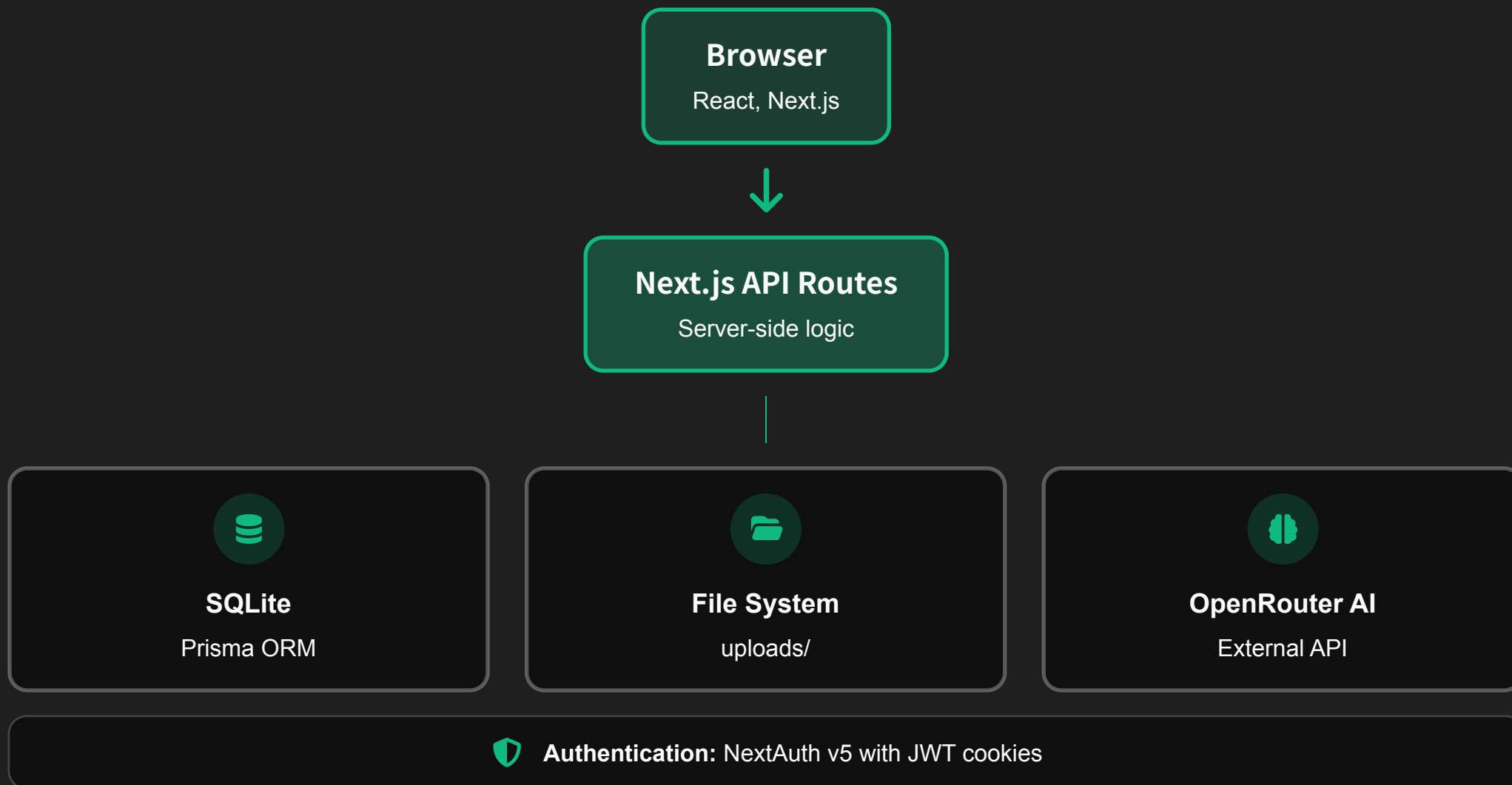


Platform

- ✓ User authentication (register/login)
- ✓ Progress tracking (auto-saved)
- ✓ Notes and highlights per page
- ✓ AI chat assistant
- ✓ Reading statistics dashboard
- ✓ Rename books, favorite, search



System Architecture





Technology Stack

Layer	Technology
Framework	Next.js 15 (App Router)
UI	React 19, Tailwind CSS 4
Language	TypeScript
Database	SQLite + Prisma ORM
Auth	NextAuth v5 (Credentials + bcrypt)
PDF Render	pdfjs-dist + react-pdf
EPUB Parser	JSZip + @xmldom/xmldom (custom)
AI	OpenRouter API
Storage	Local filesystem (uploads/)



Key Technical Challenges

01

Custom EPUB Parser

Built from scratch: container.xml → content.opf → spine → XHTML chapters. No existing library worked with Next.js server environment

02

Image Inlining

EPUB images are internal ZIP files — converted to base64 data URIs for browser display

03

Progress Auto-Save

Debounced 800ms + keepalive fetch on tab close — no progress is ever lost

04

AI Title Cleanup

"the_great_gatsby_scan_2019_fixed.pdf" → "The Great Gatsby" automatically

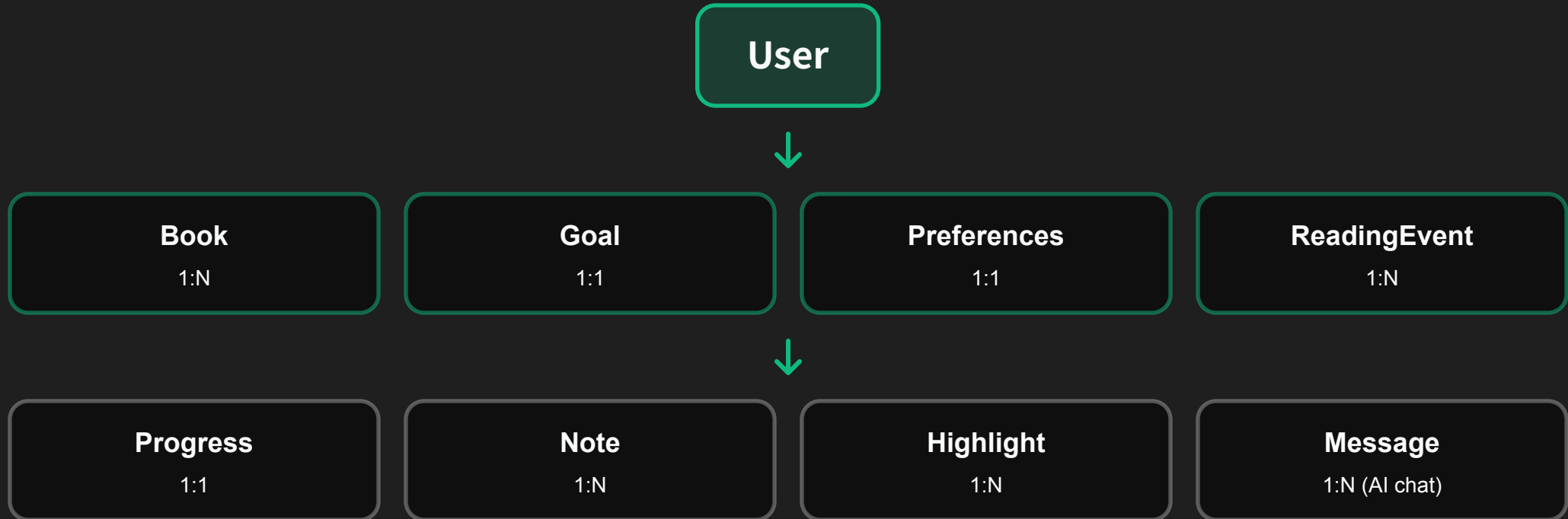
05

Toolbar Hysteresis

Reader toolbar hides only after 80px of sustained scroll — no annoying flickering



Database Schema





Results



Fully Working App

Fully working web application deployed locally



Dual Format Support

Supports PDF and EPUB reading



AI Integration

AI assistant integrated for intelligent Q&A



Complete Feature Set

Notes, highlights, statistics, goals all functional



Theme Support

Light/dark theme toggle throughout the app



Polished UI

Professional UI with smooth animations



Thank You

Bookstore — read smarter, not harder.



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